Finite-Source Cosmology from Observer-Patch Holography

Source, Transfer, and Likelihood Boundaries

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Abstract

We give a conditional source-screen construction for Observer-Patch Holography (OPH). A supplied primitive collar law fixes the angular amplitude, and an edge-center generator with its orientation-half identity fixes the tilt. Physical dilation and radial cross-covariance tomography provide two sufficient uniqueness routes beyond one-shell data. The construction states the source, covariant stress, physical scale, Boltzmann transfer and likelihood conditions needed to convert these screen quantities into cosmological observables.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The same source premises govern the dark-sector, cosmological-vacuum, and compact OPH constructions.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = \left(g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B}\right)_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \leq s \leq 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. Two sufficient routes for a source-derived lift are a scale-natural embedding that transports refinement to physical log-wavenumber dilation and complete radial cross-covariances supporting spherical-Hankel tomography. Each route has its own hypotheses. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The source Gram construction supplies a three-dimensional spatial carrier and, after adjoining a real axis, an algebraic Lorentzian carrier. A nonzero closed convex pointed displacement cone is forced to be future or past Lorentzian if the proper icosahedral rotations and all rapidities of one boost axis preserve it. This is an operational condition on histories, interventions and clock readouts; an algebraic frame action alone does not satisfy it. A time orientation fixes the future branch [1].

A conservative Fibonacci-record family realizes a controlled flat $1+3$ limit through a specified population, complete shrinking-radius neighbour reads and layer duration. Its normalized event counts converge to spacetime volume, and interior timelike intervals have ordering fraction tending to $1/10$. Given complete authenticated ancestry and interval anchors, an observer can read $(|I|/|J|)^{1/4}$ relative to a nondegenerate interval J in the same history. It converges to the proper-time ratio without using coordinates, timestamps or density in the readout. The population and local read law remain hypotheses of the limit, and the reference interval fixes only a relative time unit [1].

Cosmology requires a common physical realization of those events and clocks with expanding geometry, conserved stress, a primordial source, a radial lift and photon transfer. The finite tensor theorem uses a separately supplied $3+1$ tensor interface, nine algebraic null balances, symmetry, discrete conservation and connectedness to obtain an Einstein-shaped identity with one constant metric term. It does not identify its tensor-coordinate differences with source-event displacements. Smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. The flat count-clock construction does not supply these cosmological attachments or a likelihood.

2.1 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r}\left[\frac{1}{3}\log\frac{J_{X,r}}{\bar{J}_{X,r}}\right]. \tag{2}$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r}\mathbb{E}_{\nu_r^{\mathrm{rel}}}[q_r^{\mathsf{T}} M_r K_{\theta,r} q_r] = \frac{2E_{q,r}^{\mathrm{src}}}{d_r}. \tag{3}$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2}+1/4}+3/2+\theta/2)}{\Gamma(\sqrt{-\Delta_{S^2}+1/4}-1/2-\theta/2)}, \tag{4}$$

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}, \qquad \ell \geq 2. \tag{5}$$

A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1-\frac{P_\star}{48}, \qquad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star-\varphi)}. \tag{6}$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\text{src},r} U_r^*. \tag{7}$$

Strong finite-to-continuum convergence, a uniform covariance norm bound, and a vanishing operator residual then give

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \tag{8}$$

This identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}. \tag{9}$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

3 Dark-Sector Interface

The dark-matter paper proposes anomalous modular charge on overlap collars. For a conserved total charge Q_A in a comoving region of physical volume $V_0 a^3$, its homogeneous density satisfies

$$\rho_A = \frac{Q_A}{V_0 a^3} \propto a^{-3}. \tag{10}$$

Under the separate homogeneous continuity equation with no exchange, this dilution is the pressureless background law. Deep-regime scale covariance and quadrature composition characterize the static spherical acceleration law and baryonic Tully–Fisher scaling.

Equation (10) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that supplies these quantities from one common source and obeys the full energy-momentum ledger. That completion is not supplied here.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

The conditional source theorem selects $n_s = 1 - P_\star/48$ from the infinitesimal edge-center generator receipt. Physical primordial use requires one finite source construction that supplies this receipt together with the release law, stress, clock, freezeout, mode, scale, and radial receipts. Numerical agreement with a measured summary does not supply the missing source map.

5 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)). \tag{11}$$

All outputs must arise from one compatible background and perturbation system.

6 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. These computations supply neither a prospectively fixed cosmological comparison nor a falsification verdict.

7 Theorem scope

Object	Classification
Finite observer-screen carriers	Conditional finite construction
Screen scalar covariance	Conditional theorem
Source-functional screen amplitude and edge-center tilt	Conditional theorem; finite receipt not supplied
Radial-lift mathematics	Conditional theorem packet; one-shell unrestricted inversion is impossible
Physical source-dilation or radial-tomography receipt	Not supplied
Primordial covariant source	Not supplied
Repair-charge cosmological abundance	Not supplied
Relativistic repair stress and slip	Not supplied
Boltzmann initial-state bridge	Not supplied
Joint CMB, baryon acoustic oscillation, lensing, growth, and cluster likelihood	Not supplied
Prospectively fixed cosmological comparison	Not supplied by these diagnostics

8 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are not supplied here. Existing cosmological calculations are diagnostics and carry no falsification verdict.

References

[1] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.tex.

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